



**PHYSICS
STANDARD LEVEL
PAPER 1**

Topic A - Mechanics Study Guide

May 2009 - May 2013

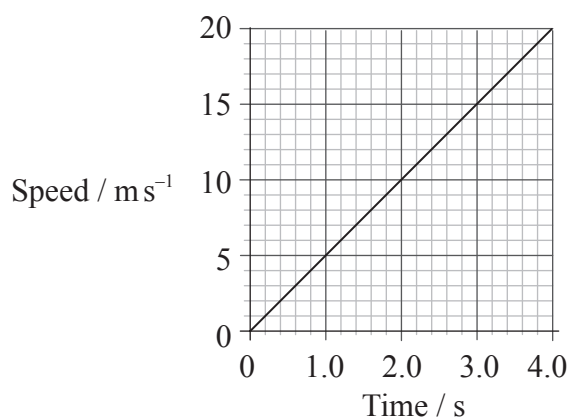
45 minutes

INSTRUCTIONS TO CANDIDATES

- Do not open this examination paper until instructed to do so.
- Answer all the questions.
- For each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.



3. A skydiver jumped out of an airplane. On reaching a terminal speed of 60 m s^{-1} , she opened her parachute. Which of the following describes her motion after opening her parachute?
- A. She went upwards for a short time, before falling to Earth at a speed of 60 m s^{-1} .
 - B. She continued downwards at 60 m s^{-1} , but hit the ground with less force.
 - C. She continued to fall but reached a new terminal speed of less than 60 m s^{-1} .
 - D. She went upwards for a short time, before falling to Earth at a speed of less than 60 m s^{-1} .
4. The graph is a speed versus time graph for an object that is moving in a straight line.



The distance travelled by the object during the first 4.0 seconds is

- A. 80 m.
- B. 40 m.
- C. 20 m.
- D. 5 m.

5. The diagram shows a girl attempting (but failing) to lift a heavy suitcase of weight W . The magnitude of the vertical upwards pull of the girl on the suitcase is P and the magnitude of the vertical reaction of the floor on the suitcase is R .

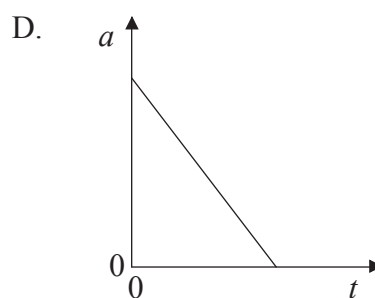
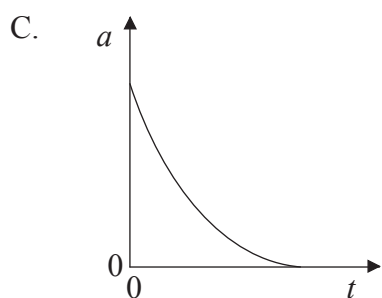
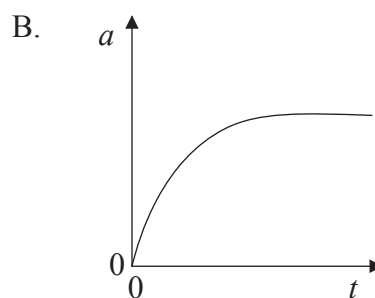
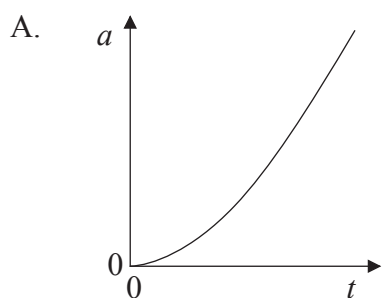


Which equation correctly relates W , P and R ?

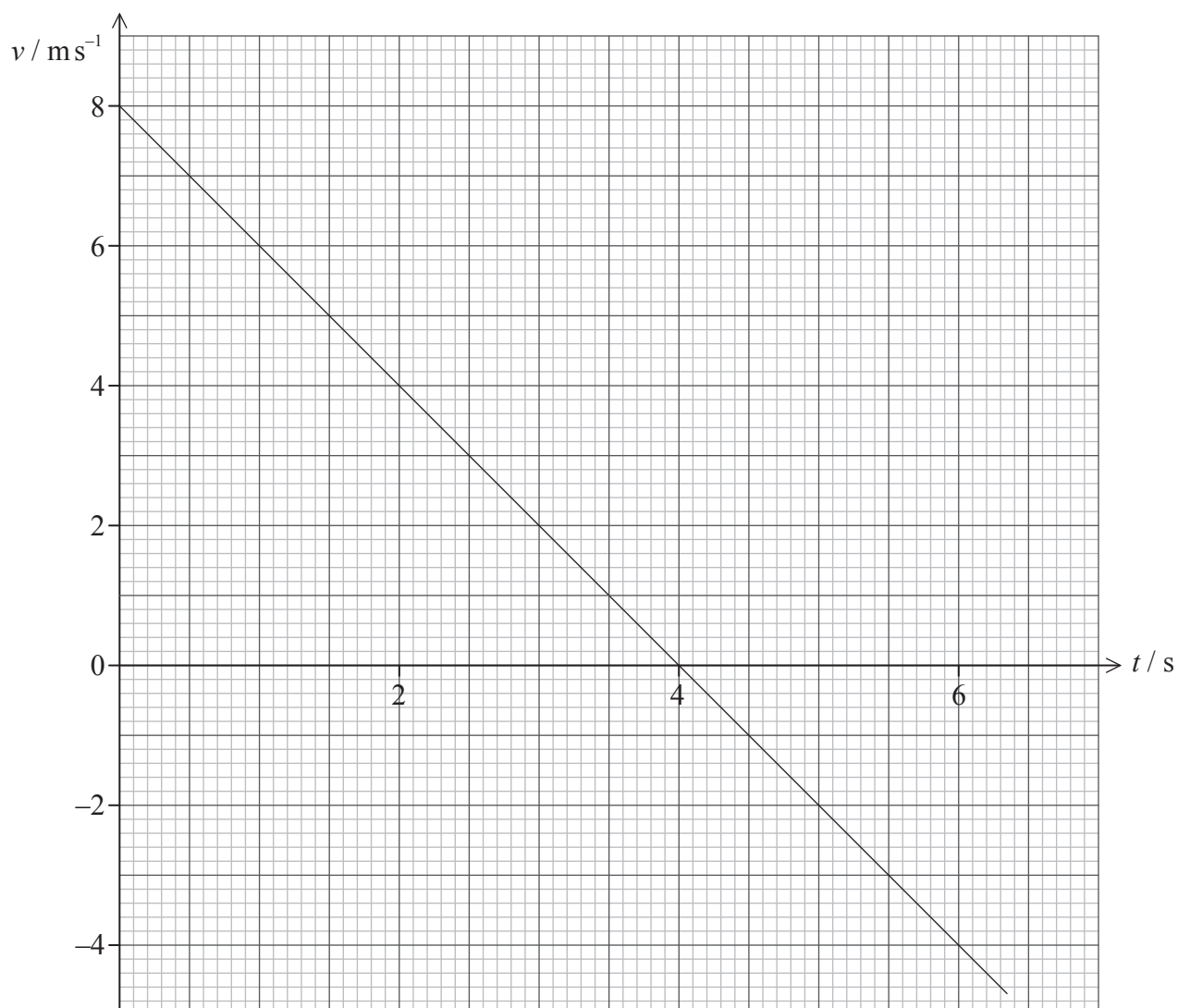
- A. $W = P + R$
 - B. $W > P + R$
 - C. $W < P + R$
 - D. $W = P = R$
6. Objects A and B collide together. They end up joined together and stationary. During the collision, a force $+F$ is exerted on object A by object B . According to Newton's third law, there will also be a force of
- A. $-F$ acting on object B .
 - B. $-F$ acting on object A .
 - C. $+F$ acting on object B .
 - D. $+F$ acting on object A .

7. A lift (elevator) is operated by an electric motor. It moves between the 10th floor and the 2nd floor at a constant speed. One main energy transformation during this journey is
- A. gravitational potential energy → kinetic energy.
 - B. electrical energy → kinetic energy.
 - C. kinetic energy → thermal energy.
 - D. electrical energy → thermal energy.

3. A ball, initially at rest, is dropped in the air from a great height. Air resistance is not negligible. Which of the following graphs best shows the variation with time t of the acceleration a of the ball?



4. The graph below shows the variation with time t of the velocity v of an object moving along a straight line.



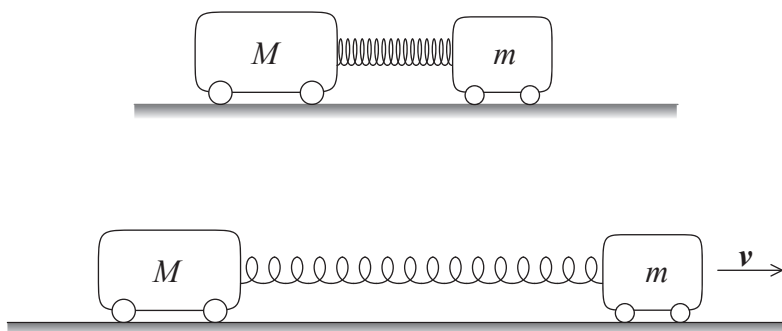
The displacement of the object between $t=0\text{ s}$ and $t=6.0\text{ s}$ is

- A. 2.0 m.
- B. 12 m.
- C. 20 m.
- D. 24 m.

5. A constant force of magnitude F is applied to a mass m for a time interval Δt . The magnitude of the impulse given to the mass equals

- A. $\frac{F}{m}$.
- B. $\frac{F}{\Delta t}$.
- C. $F\Delta t$.
- D. $\frac{F\Delta t}{m}$.

6. Two carts of different mass m and M are connected by a spring. They are pushed together such that the spring is compressed.

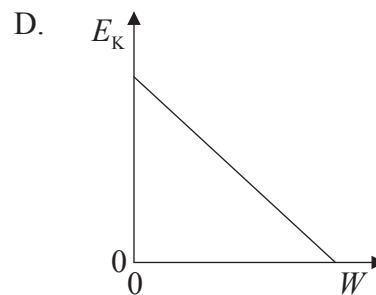
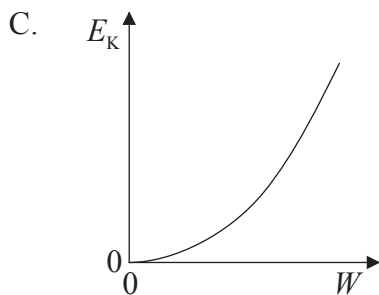
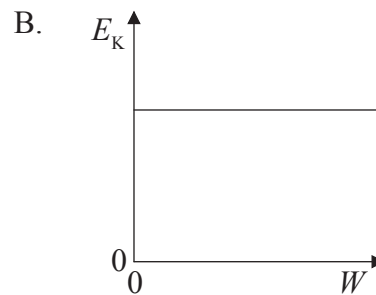
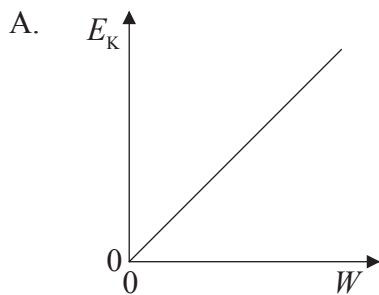


After the carts are released, the cart of mass m moves with velocity v . The change in the momentum of mass M is

- A. mv .
- B. $-mv$.
- C. Mv .
- D. $-Mv$.

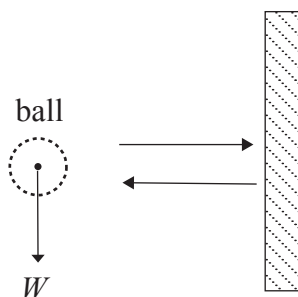
7. A lamp of weight W is suspended by a wire fixed to the ceiling. With reference to Newton's third law of motion, the force that is equal and opposite to W is the
- A. tension in the wire.
 - B. force applied by the ceiling.
 - C. force exerted by the lamp on the Earth.
 - D. force exerted by the Earth on the lamp.

8. A constant force acts on a mass that is initially at rest. Which of the following graphs best shows how the kinetic energy E_k of the mass changes with the work W done on the mass? Friction is negligible.



3. The time taken for a stone dropped from rest to fall vertically through 16 m is 2.0 s. Based on these measurements, what is the best estimate for the acceleration of free fall?
- A. 4.0 m s^{-2}
 - B. 8.0 m s^{-2}
 - C. 9.8 m s^{-2}
 - D. 10 m s^{-2}
4. A wooden block is sliding down an inclined plane at constant speed. The magnitude of the frictional force between the block and the plane is equal to
- A. zero.
 - B. the magnitude of the weight of the block.
 - C. the magnitude of the component of weight of the block parallel to the plane.
 - D. the magnitude of the component of the normal reaction parallel to the plane.

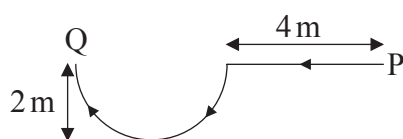
5. Which of the following is a correct statement of Newton's second law of motion?
- A. A force acting on a body is proportional to the mass of the body.
 - B. The rate of change of momentum of a body is equal to the net external force acting on the body.
 - C. The momentum of a body is proportional to the net external force acting on the body.
 - D. A force acting on a body is proportional to the acceleration of the body.
6. A ball of weight W is travelling horizontally towards a vertical wall. It strikes the wall and rebounds horizontally. The change in the magnitude of the momentum of the ball is Δp . Which of the following is the magnitude of the impulse that the ball imparts to the wall?



- A. $W + \Delta p$
 - B. $W - \Delta p$
 - C. W
 - D. Δp
7. Two objects undergo an inelastic collision. Which of the following is correct in respect of both the conservation of momentum and the conservation of total energy of the system?

	Momentum	Total energy
A.	conserved	not conserved
B.	conserved	conserved
C.	not conserved	not conserved
D.	not conserved	conserved

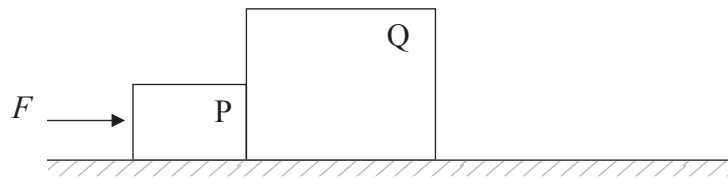
3. Samantha walks along a horizontal path in the direction shown. The curved part of the path is a semi-circle.



The magnitude of her displacement from point P to point Q is approximately

- A. 2 m.
- B. 4 m.
- C. 6 m.
- D. 8 m.

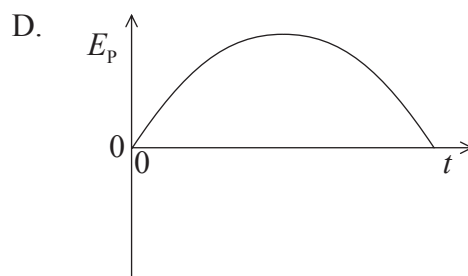
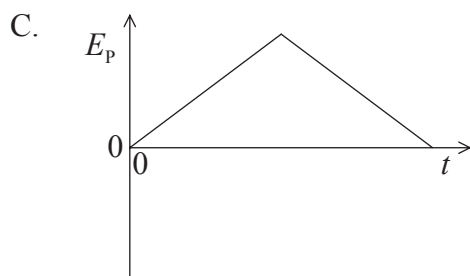
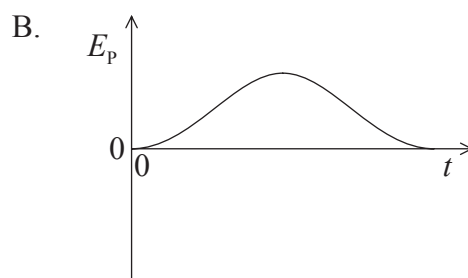
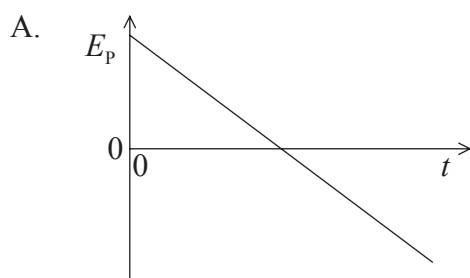
4. Which of the following may be determined from a speed-time graph?
- A. Displacement
 - B. Distance
 - C. Power
 - D. Force
5. Stephen pushes two boxes P and Q, that stay in contact, along a rough table, with a force F of 30 N. Box P has a mass of 2.0 kg and box Q has a mass of 4.0 kg. Both boxes move with constant speed.



The resultant force on box Q is

- A. 0 N.
- B. 5.0 N.
- C. 15 N.
- D. 30 N.

7. A ball is thrown vertically upwards and comes down again. Air resistance is negligible. Which of the following graphs shows how the gravitational potential energy E_p varies with time t ?



8. A pump extracts water from a well of depth h at a constant rate of $R \text{ kg s}^{-1}$. What is the power required to raise the water?

A. $\frac{R}{gh}$

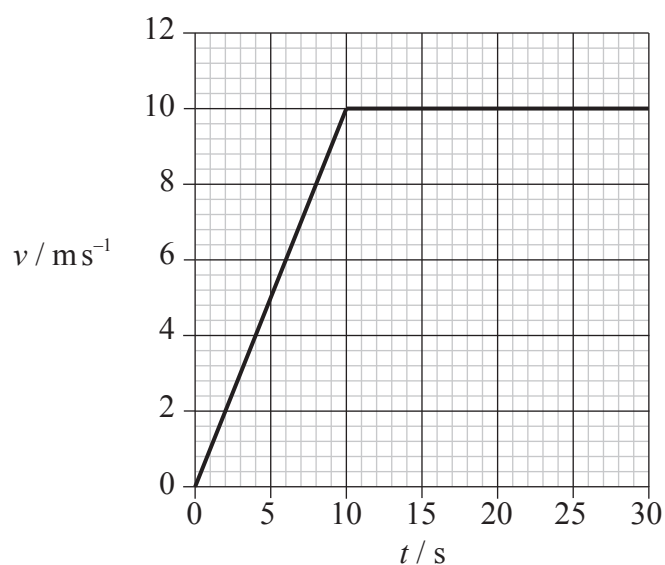
B. Rgh

C. $\frac{Rg}{h}$

D. $\frac{hg}{R}$

3. A skydiver of mass 80 kg falls vertically with a constant speed of 50 ms^{-1} . The upward force acting on the skydiver is approximately
- A. 0 N.
 - B. 80 N.
 - C. 800 N.
 - D. 4000 N.

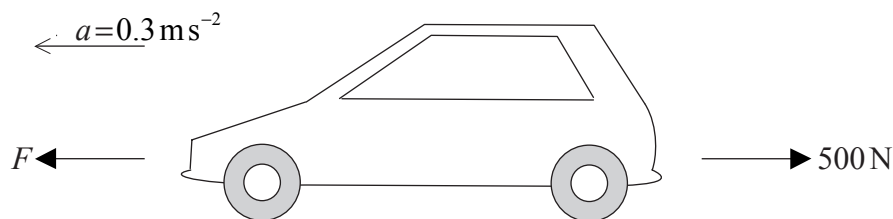
4. Joseph runs along a long straight track. The variation of his speed v with time t is shown below.



After 25 seconds Joseph has run 200 m. Which of the following is correct at 25 seconds?

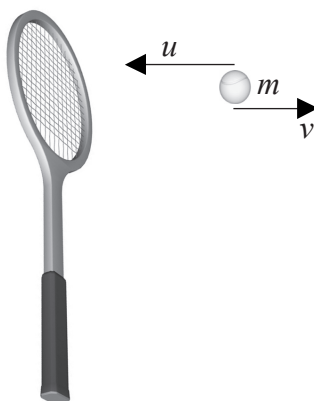
	Instantaneous speed / m s^{-1}	Average speed / m s^{-1}
A.	8 m s^{-1}	8 m s^{-1}
B.	8 m s^{-1}	10 m s^{-1}
C.	10 m s^{-1}	8 m s^{-1}
D.	10 m s^{-1}	10 m s^{-1}

5. A car of mass 1000 kg accelerates on a straight, flat, horizontal road with an acceleration $a = 0.3\text{ ms}^{-2}$. The driving force F on the car is opposed by a resistive force of 500 N .



The net (resultant) force on the car is

- A. 200 N .
 - B. 300 N .
 - C. 500 N .
 - D. 800 N .
6. A tennis ball of mass m moving horizontally with speed u strikes a vertical tennis racket. The ball bounces back with a horizontal speed v .



The magnitude of the change in momentum of the ball is

- A. $m(u + v)$.
- B. $m(u - v)$.
- C. $m(v - u)$.
- D. zero.

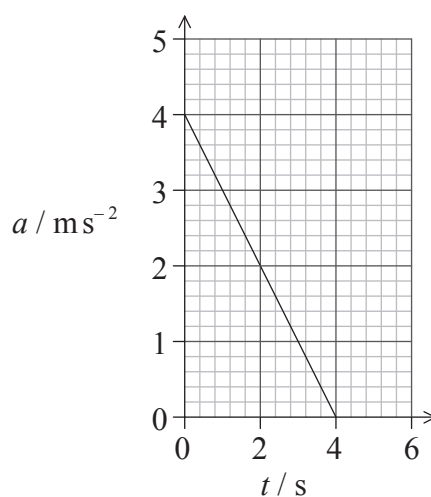
7. A brother and sister take the same time to run up a set of steps. The sister has a greater mass than her brother. Which of the following is correct?

	Has done the most work	Has developed the greatest power
A.	brother	brother
B.	brother	sister
C.	sister	brother
D.	sister	sister

8. A nuclear power station produces 10 GW of electrical power. The power generated by the nuclear reactions in the core of the reactor is 25 GW. The efficiency of the power station is

- A. 15 %.
- B. 35 %.
- C. 40 %.
- D. 60 %.

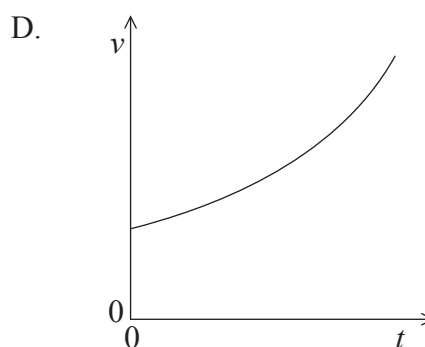
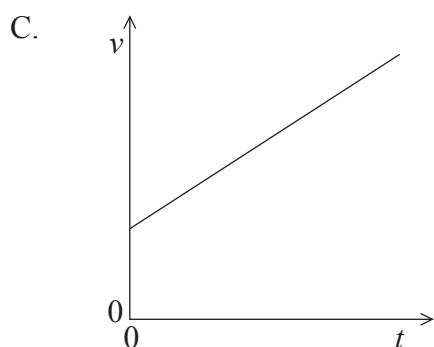
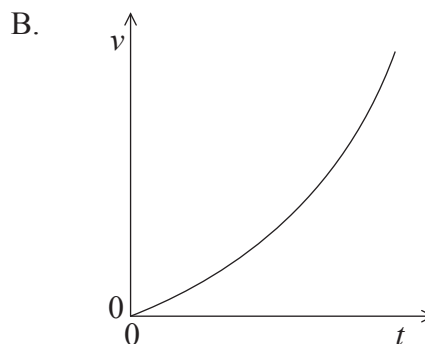
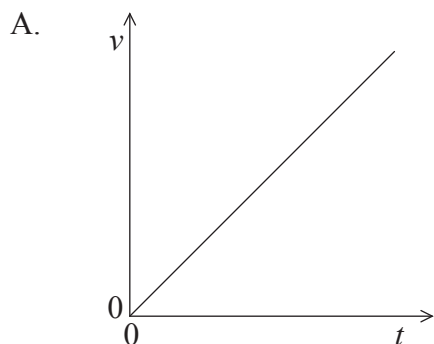
4. The graph shows the variation with time t of the acceleration a of an object.



Which of the following is the change in velocity of the object in the time interval 0 to 4 s?

- A. -8 m/s^{-1}
- B. -4 m/s^{-1}
- C. $+4 \text{ m/s}^{-1}$
- D. $+8 \text{ m/s}^{-1}$

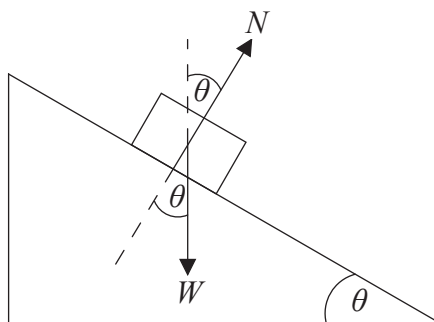
5. A car accelerates from rest. The acceleration increases with time. Which graph shows the variation with time t of the speed v of the car?



6. Which of the following is the condition for a body to be in translational equilibrium?

- A. The resultant force on the body in any direction is zero.
- B. The velocity of the body in any direction is zero.
- C. No external force is acting on the body.
- D. No work is done on the body.

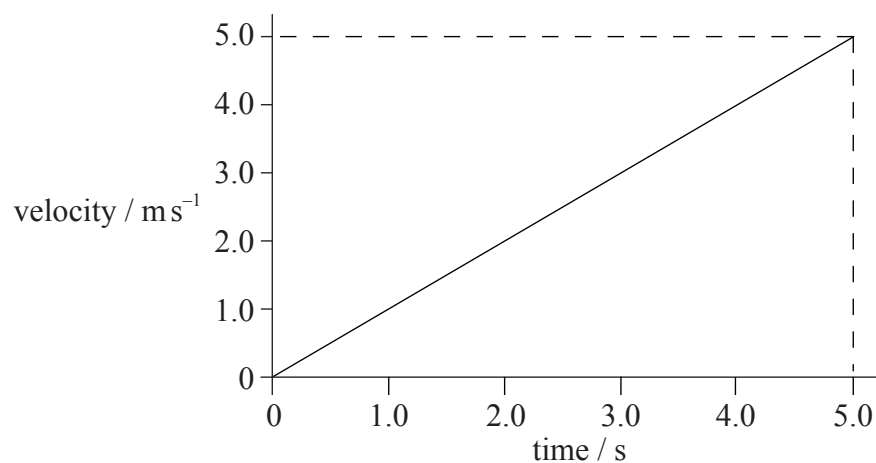
2. The diagram below shows the forces acting on a block of weight W as it slides down a slope. The angle between the slope and the horizontal is θ , the normal reaction force on the block from the slope is N and friction is negligible.



Which of the following gives the resultant force on the block?

- A. $W \sin \theta$
- B. $W \cos \theta$
- C. $N \sin \theta$
- D. $N \cos \theta$

3. The velocity–time graph for an accelerating object that is travelling in a straight line is shown below.

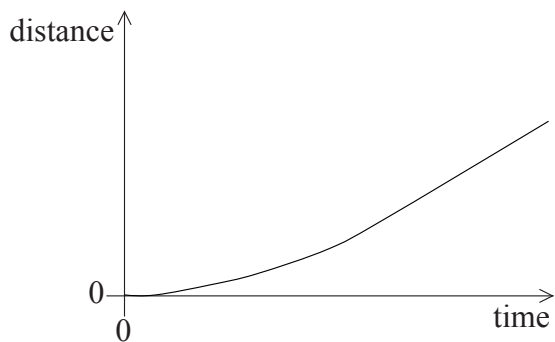


Which of the following is the change in displacement of the object in the first 5.0 seconds?

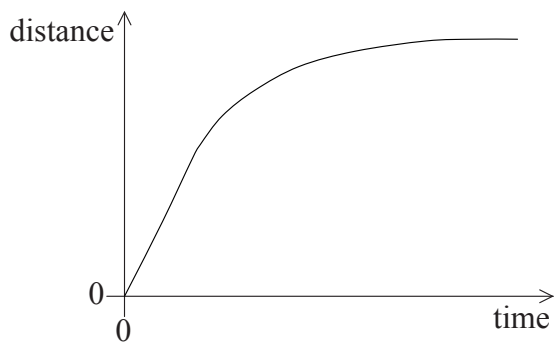
- A. 25.0 m
- B. 12.5 m
- C. 5.0 m
- D. 1.0 m

4. An object falls vertically from rest. Air resistance acts on the object and it reaches a terminal speed. Which of the following is the distance–time graph for its motion?

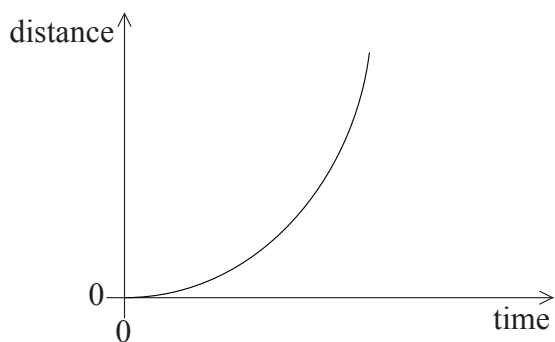
A.



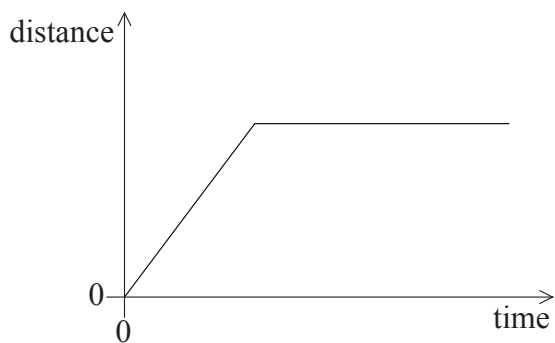
B.



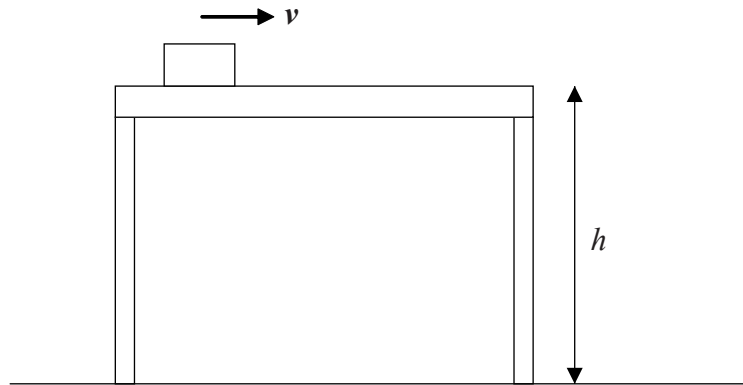
C.



D.



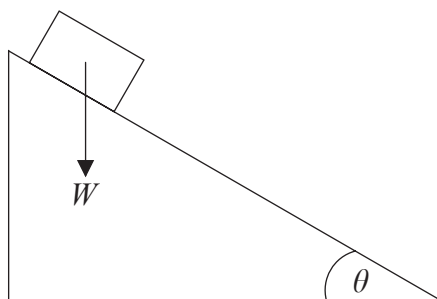
6. A block of mass m is moving at constant velocity v along a frictionless surface that is height h above the ground.



Which expression gives the work necessary to maintain the constant velocity?

- A. mgh
 - B. $\frac{1}{2}mv^2$
 - C. $mgh + \frac{1}{2}mv^2$
 - D. zero
7. Which of the following is an elastic collision?
- A. Two railway trucks collide and they link together.
 - B. Two railway trucks collide and they do not link together.
 - C. Two gas molecules collide and each changes direction.
 - D. Two gas molecules collide and a bond is formed between them.

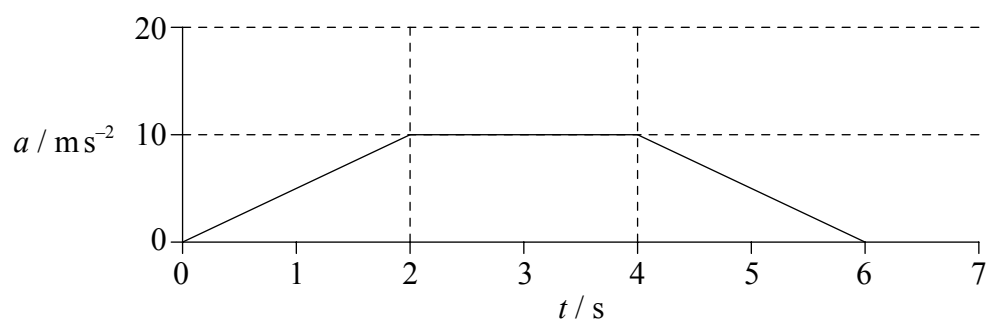
2. An object slides down an inclined plane that makes an angle θ with the horizontal. The weight of the object is W .



Which of the following is the magnitude of the component of the weight parallel to the plane?

- A. $W \sin \theta$
- B. $\frac{W}{\sin \theta}$
- C. $W \cos \theta$
- D. $\frac{W}{\cos \theta}$

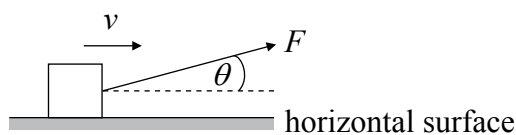
3. The graph shows the acceleration a of an object as time t varies.



What is the magnitude of the change in the velocity of the object between 0 and 3 seconds?

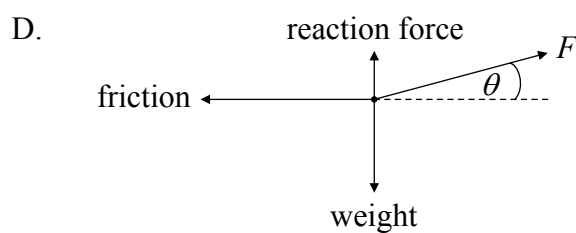
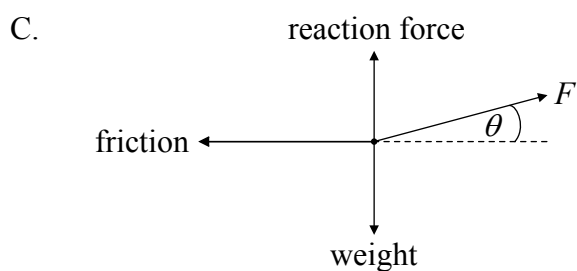
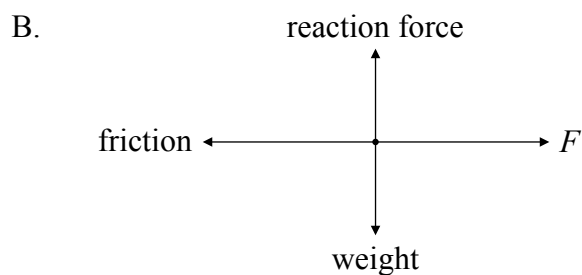
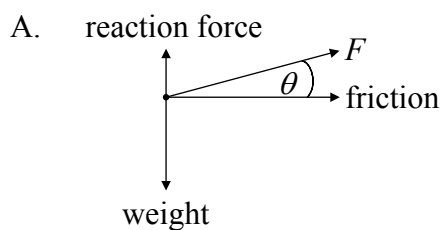
- A. 5 ms^{-1}
- B. 10 ms^{-1}
- C. 20 ms^{-1}
- D. 30 ms^{-1}

4. A force F acts on a block at an angle θ with respect to a horizontal surface.



The block is moving with a constant velocity v along the surface. A resistive force acts on the block.

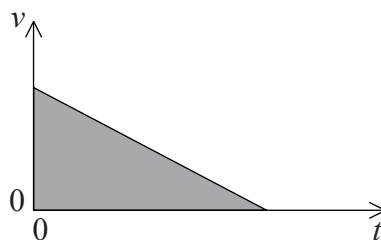
Which of the following correctly represents the forces acting on the block?



5. The momentum of a particle stays constant provided that

- A. it moves in a circle with constant speed.
- B. its acceleration is uniform.
- C. the net internal force acting on it is zero.
- D. the net external force acting on it is zero.

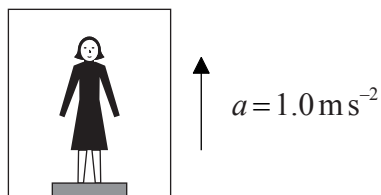
3. The graph below shows how velocity v varies with time t for a ball thrown vertically upwards from the Earth's surface.



The shaded area is equal to the

- A. displacement.
 - B. final velocity.
 - C. change in velocity.
 - D. acceleration.
4. Two identical balls are dropped from a tall building, one a few seconds after the other. Air resistance is **not** negligible. As the balls fall, the distance between the balls will
- A. decrease.
 - B. increase.
 - C. increase then remain constant.
 - D. remain constant.
5. Which of the following is always true for an object moving in a straight line at constant speed?
- A. No forces act on the object.
 - B. No resultant force acts on the object.
 - C. The momentum of the object is zero.
 - D. No work is being done on the object.

6. A person of weight 600 N is standing on a weighing scale in a lift (elevator). The lift is accelerating upwards at 1.0 m s^{-2} . Which of the following is the reading on the scale?



- A. 0 N
- B. 540 N
- C. 600 N
- D. 660 N
7. A spaceship is moving at constant speed in outer space. An internal explosion causes it to break into two pieces that move in opposite directions. Which of the following best describes what happens to its total momentum and total kinetic energy as a result of the explosion?

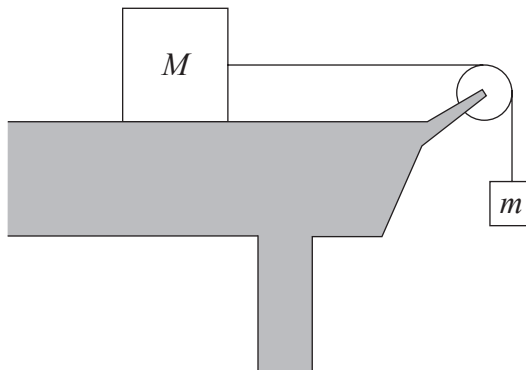
	Total momentum	Total kinetic energy
A.	unchanged	unchanged
B.	unchanged	increased
C.	increased	unchanged
D.	increased	increased

3. An object, initially at rest, travels a distance d in a time t at a constant acceleration. What is the time taken for the object to travel $16d$ from rest at the same acceleration?
- A. $16t$
 - B. $8t$
 - C. $4t$
 - D. $2t$

4. An object is released above the surface of Earth. Which of the following correctly describes the speed and acceleration before it reaches terminal speed?

	Speed	Acceleration
A.	increases	remains constant
B.	increases	decreases
C.	remains constant	remains constant
D.	remains constant	decreases

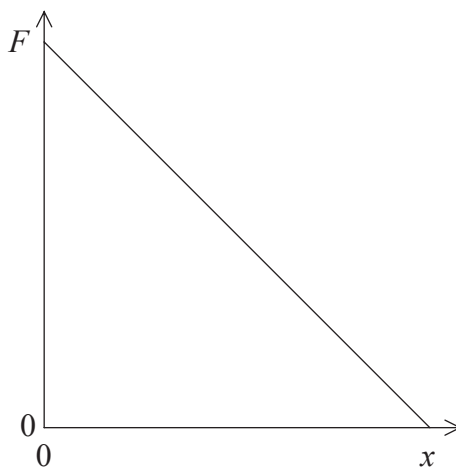
5. An object of mass m is connected via a frictionless pulley to an object of mass M , where $M > m$. M rests on a horizontal frictionless surface.



What is the acceleration of the system?

- A. $\frac{mg}{(M + m)}$
- B. $\frac{(M + m)g}{m}$
- C. $\frac{gm}{M}$
- D. Zero

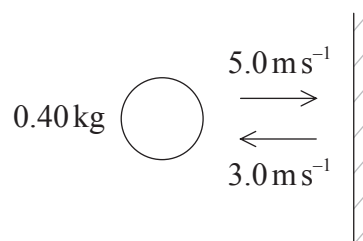
6. The graph shows the variation with distance x of the magnitude of the net force F acting on a body initially at rest.



Which of the following describes how the kinetic energy and the acceleration of the body change with distance?

	Kinetic energy	Acceleration
A.	decrease	decrease
B.	decrease	increase
C.	increase	decrease
D.	increase	increase

7. A ball of mass 0.40 kg travels horizontally and strikes a vertical wall with a speed of 5.0 ms^{-1} . It rebounds horizontally with a speed of 3.0 ms^{-1} . The ball is in contact with the wall for a time of 0.20 s .



What is the average magnitude of the force exerted by the ball on the wall?

- A. 0.16 N
- B. 0.64 N
- C. 4 N
- D. 16 N